Kewen (Kolvin) Liu

(857) 294-6549 | kewenliu.kolvin@gmail.com| LinkedIn: kolvin-liu/

EDUCATION

Georgia Institution of Technology | M.S. in Computer Science *August 2024 – Dec 2026*

- **Interest:** Accelerate AI inference from system infra and/or algorithms.

Boston University | B.A. in Statistic and Computer Science *Sep 2019 – May 2023*

- **TA**: CS330 Introduction to Analysis of Algorithm *Sep 2022 – May 2023*

TECHNICAL TOOLS

Programming languages: Python, C/C++, Go, Java, Bash, JavaScript/TypeScript, SQL, HTML/CSS, Ocaml, R
AI/ML: PyTorch, Hugging Face libraries, vLLM, Ray, CUDA, OpenMP, ONNX, OpenCV
SWE: Linux/Ubuntu, Docker, Git, Slurm, Jenkins, PyTest, MongoDB, Spark, CI/CD, Flask, Node.js

RESEARCH

- **Research project**: Improving End-to-end Training Time in RL Post-training by Branching (poster, GitHub Repo)
- **Publication:** Kewen Liu, "Analysis of Features of Different Activation Functions," 2021 2nd International Conference on Computing and Data Science (CDS), 2021, pp. 421-424, doi: 10.1109/CDS52072.2021.00078.

WORK EXPERIENCE

TileAI | ***AI Systems Intern (TileRT Team)*** *Jan 2026 – Present*

- Analyzed RL post-training frameworks (**VERL**, **SLiME**, **AREAL**) from systems perspective to identify long-tail rollout bottlenecks impacting effective GPU throughput
- Profiled large-scale LLM rollouts on **DeepSeek-V3.2** and **GLM-5.0** to quantify tail latency distribution and multi-GPU underutilization during drain phase
- Working on defining architectural blueprint for a runtime-aware RL framework leveraging TileRT backend

NVIDIA | ***Software Engineer Intern (TensorRT ModelOpt)*** *Feb 2025 – Aug 2025*

- Architected distributed CI and benchmarking pipelines for large-scale LLM quantization across multi-GPU clusters
- Performed post-training quantization on frontier foundation models (e.g., **DeepSeek-R1**, **Llama4**, **Kimi-K2**, **Qwen3**) in **FP16/FP8/FP4/INT8** or **mixed-precision** modes with ModelOpt
- Led automated PTQ integration workflows across across DL frameworks
  - Engineered PTQ workflows under **NeMo**'s framework, handling parameter mapping and config translation
  - Implemented distributed calibration for PTQ under **Megatron-Core (Megatron-LM)** training infrastructure
  - Built deployment testing pipelines using **TensorRT-LLM** or/and **vLLM** with **Ray Serve**
- Profiled quantization performance per release (with **Nsight's**, **torch.profiler**, **CUDA APIs**) to track memory usage for OOM risk and imbalance workload on multi-GPU nodes
- Refactored performance result database including: (1) optimized retrieval logic and, (2) eliminated **18%** data loss from failed/interrupted pipelines with session cleanup and backup fetches

Symposia Studio LLC | ***Backend Intern*** *Sep 2023 – Jan 2024*

- Developed web spider API and behavior prediction model for TikTok bot detection software with **95% accuracy**

Sichuan Dongfang Water Co., Ltd. | ***AI/CV Research Intern*** *May 2021 – August 2021*

- Researched and implemented vision pre-processing algorithm for river-cleaning robots with **OpenCV**

PROJECTS

CUDA Implementation of Bitonic Sort *Sep 2025*

- Developed bitonic sort in CUDA using 3-tier GPU memory optimization strategy (local/warp, shared, global), achieved **183% speedup** over CPU computation on 100 million element arrays, techniques include **vectorized memory access**, **warp shuffles**, **loop unrolling**, **dim size auto tuning**, **branch-free algorithm**, etc.

Distributed System Raft *Feb 2023 – May 2023*

- Reproduced **Raft** in Golang with integration of **MapReduce** on 7 virtual servers communicated through channel-based RPC, achieved system reliability and fault tolerance under **random server failures, network disconnections, and network partitions**, simulated by shell/bash scripts